Practice workbook · Log equations that turn into a quadratic

Solving equations that mix log_a x and log_x a (or other bases) by changing base so they become a quadratic in one log.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-11

1. Question 8 · part whole · 7 marks

2020 NOV · Paper 02 (regional R)

Worked-example retry · Attempt part whole only.

8 Solve the equation $\log_3 x - 2\log_x 3 = 1$ (7)

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1. Question 8 · part whole · 7 marks

2020 NOV · Paper 02 (regional R)

Worked-example retry · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 7 marks)

1. Question 8 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 5 · part b · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 (a) Show that $\log_4 32 = \frac{5}{2}$

(2)

(b) Hence, or otherwise, find the exact solutions of the equation

$$\log_2 x - \log_4 32 + \frac{1}{4}\log_x 16 = 0$$

(7)

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2. Question 5 · part b · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 9 marks)

2. Question 5 · part b · 7 marks · Working and answer

Working and answer

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 A curve C has equation $y = \log_{10}(x + 2)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes.

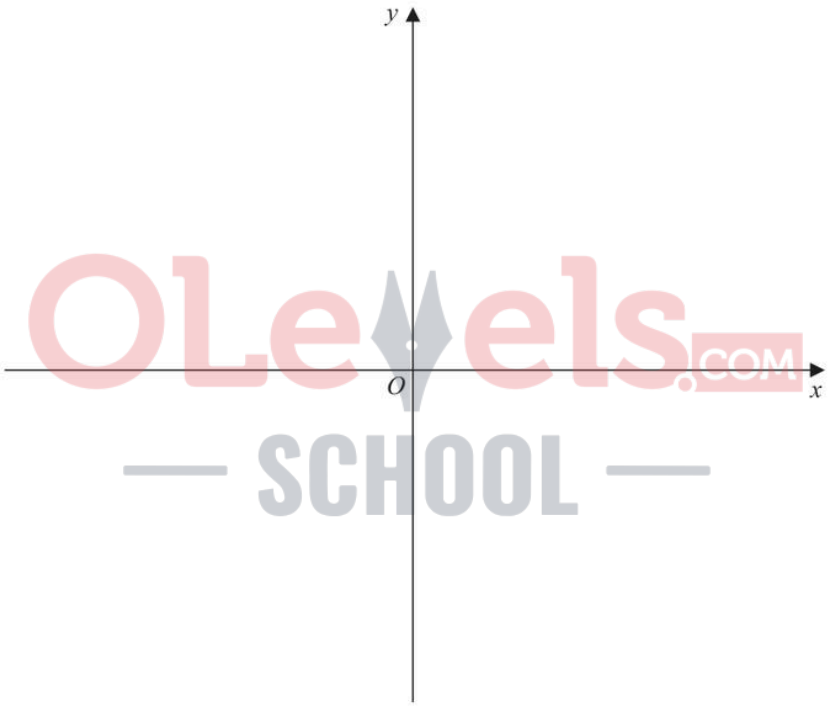
(2)

(b) Solve the equation $2(\log_a 4 + \log_a 16) = 1$

(3)

(c) Solve the equation $5\log_q 16 + 4\log_2 q = 24$

(6)



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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 7 continued

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(Total for Question 7 is 11 marks)

3. Question 7 · part c · 6 marks · Working and answer

Working and answer

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4. Question 6 · part whole · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

6 Solve the equation

$$\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = 0$$

giving your answers to 3 significant figures.

(8)

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4. Question 6 · part whole · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

4. Question 6 · part whole · 8 marks · Working and answer

Working and answer

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5. Question 7 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

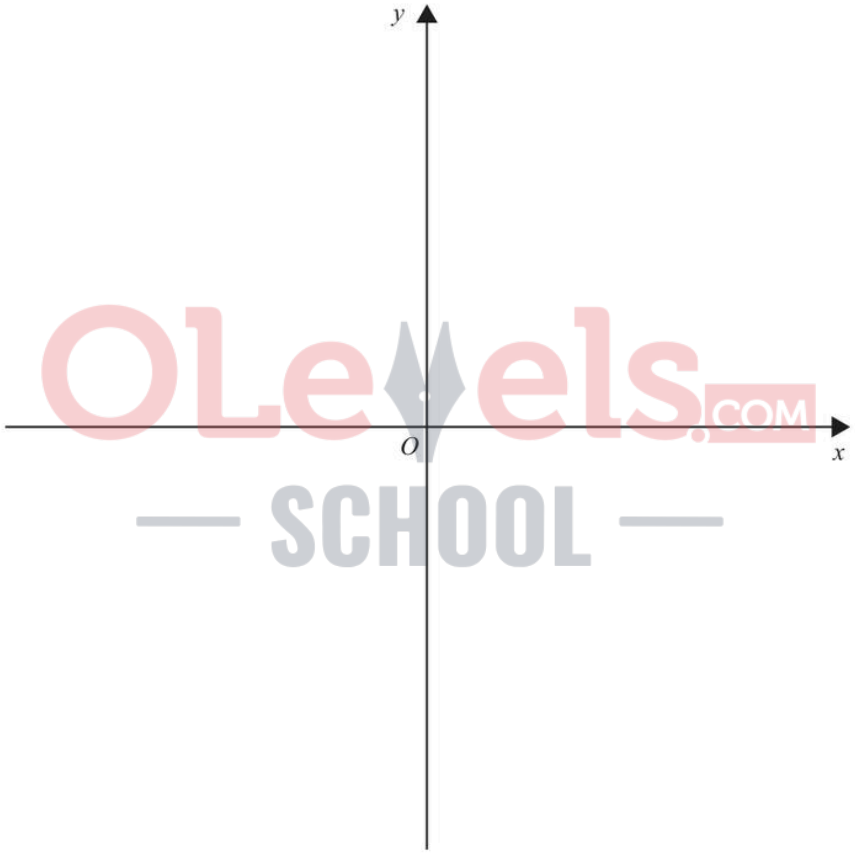
7 The curve C has equation $y = -\log_4(x + 4)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes and the equation of any asymptote to C.

(4)

(b) Solve the equation $\log_{(x+4)} 256 - \log_4(x + 4) = 0$

(5)



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5. Question 7 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

5. Question 7 · part b · 5 marks · Working and answer

Working and answer

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6. Question 9 · part ii · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

9 (i) Solve the equation $3(\log_a 9 + \log_a 27) = 1$

Give your answer in the form $a = b^c$ where b is prime and c is an integer.

(3)

(ii) Solve the equation $\log_4 p + \log_p 256 = -4$

(6)

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6. Question 9 · part ii · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

Question 9 continued

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(Total for Question 9 is 9 marks)

6. Question 9 · part ii · 6 marks · Working and answer

Working and answer

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7. Question 6 · part ii · 7 marks

2024 NOV · Paper 02

Further practice · Attempt part ii only.

6 (i) Solve the equation $5(\log_b 9 + \log_b 3) = 3$

(4)

(ii) Solve the equation $3\log_3 x + 3\log_x 27 = 8\log_4 128$
Give your answers in exact form.

(7)

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7. Question 6 · part ii · 7 marks

2024 NOV · Paper 02

Further practice · Attempt part ii only.

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Question 6 continued

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(Total for Question 6 is 11 marks)

7. Question 6 · part ii · 7 marks · Working and answer

Working and answer

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8. Question 10 · part whole · 7 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

10 Solve the equation

$$\log_4 x^3 + 8\log_x 64 = 22$$

(7)

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8. Question 10 · part whole · 7 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 10 continued

(Total for Question 10 is 7 marks)

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8. Question 10 · part whole · 7 marks · Working and answer

Working and answer

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